

Example 1. The arrival process of **returning** and **new** customers to a shop by **t** hours after opening are independent Poisson process.

Returning customers arrive at a mean rate of **2** per hour, and new customers arrive at a mean rate of **1** per hour.

Let **T** be the time (in hours) of the first arrival of a customer (returning or new).

$\mathbb{E}[T \mid \text{no customers have arrived in the first half hour}]$

A. Principle of Thinning. Suppose that the arrival of vehicular crashes on a highway is a Poisson process $\{Z_t\}_{t \geq 0}$ with rate λ .

A given vehicular crash, independent of all else:

→ involves a motorcycle with probability p

→ does not involve a motorcycle with probability $q = 1 - p$

Let:

→ $\{X_t\}_{t \geq 0}$ be the counting process of crashes:

→ $\{Y_t\}_{t \geq 0}$ be the counting process of crashes:

Then, given positive time t , the joint mass function of X_t and Y_t is:

$$p(m, n) = \mathbb{P}(X_t = m, Y_t = n) =$$

Principle of Thinning. If, in a Poisson process with rate parameter λ , arrivals are, independently of all else, labelled “A” with probability p and labelled “B” with probability $q = 1 - p$, then the arrival processes of A’s and B’s are **independent** Poisson process with rate parameters:

$$\lambda_A =$$

$$\lambda_B =$$

As a consequence:

$$\lambda =$$

$$p =$$

$$q =$$

Note: This idea can be extended if we allow more independent labels, say labels “ A_i ” with probability p_i , in which case the parameter is $\lambda_{A_i} = \lambda p_i$.

Example 2. Use the principle of thinning (not integrals) to show that, if S and T are independent exponential random variables with parameter λ and μ , then:

$$\mathbb{P}(S < T) = \frac{\lambda}{\lambda + \mu}$$

Example 3. Passengers arrive at a bus stop according to a Poisson process with rate λ . The bus arrives at time $T \sim \text{Exp}(\mu)$, independently of the passengers. Let N be the number of passengers that make it on the bus.

Find the distribution of N using principles of superposition and thinning.